בס"ד

מיקרו-מעבדים ואסמבלייר – שיעורי בית 1

הערה:

מגישים-

נתן קרומביין, ת"ז: 341004471

אורי מימון, ת"ז: 216383174

מהלך המטלה:

במטלה זו התבקשנו בגדול לשמור את תעודות הזהות שלנו בשני מערכים, וכן לשמור במערך נפרד את כתובות המערכים הראשונים. ולאחר זאת להדפיס למסך את מספרי הזהות בצבעים שונים.

ראשית, הגדרנו שנרצה לעבוד עם שלושה מערכים וכן שמרנו סיגמנטים עבור ה-data וכן עבור ה-screen:

```
.model small
.stack 100h
.data
    ID1 db 9d dup (?) ; Initialize an array that will save the first ID
    ID2 db 9d dup (?) ; Initialize an array that will save the secend ID
    DPOINTERS dw 2d dup (?) ; Initialize an array that will save the pointers to the first two arrays
.code

START:
    ; Initializing data segment
    mov ax, @data
    mov ds, ax

    ; Setting extra segment to screen memory
    mov ax, 0b800h
    mov es, ax
```

לאחר מכן, שמרנו בתוך DPOINTERS את כתובות שני המערכים האחרים:

```
    ; Setting DPOINTERS with ID1, ID2 addresses
    mov bx, offset DPOINTERS ; Sets bx to stock the offset of DPOINTERS address
    mov [bx], offset ID1 ; Sets the first place in DPOINTERS to be ID1's adderess
    mov [bx+2], offset ID2 ; Sets the second place in DPOINTERS to be ID2's adderess

```

מילאנו את מערכי תעודות הזהות בערכים הנכונים:

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```
; Setting first ID value
mov bp, WORD [bx] ; Set bp to stock up ID1's adderess
mov ds:[bp],51 ; Puts in the 1th place in ID1 array the 1th digit - '3'.
mov ds:[bp+1],52 ; Puts in the 2th place in ID1 array the 2th digit - '4'.
mov ds:[bp+2],49 ; Puts in the 3th place in ID1 array the 3th digit - '1'.
mov ds:[bp+3],48 ; Puts in the 4th place in ID1 array the 4th digit - '0'.
mov ds:[bp+4],48 ; Puts in the 5th place in ID1 array the 5th digit - '0'.
mov ds:[bp+5],52 ; Puts in the 6th place in ID1 array the 6th digit - '4'.
mov ds:[bp+6],52 ; Puts in the 7th place in ID1 array the 7th digit - '4'.
mov ds:[bp+7],55 ; Puts in the 8th place in ID1 array the 8th digit - '7'.
mov ds:[bp+8],49 ; Puts in the 9th place in ID1 array the 9th digit - '1'.

; Setting second ID value
mov bp, WORD [bx+2] ; Set bp to stock up ID2's adderess
mov ds:[bp],50 ; Puts in the 1th place in ID1 array the 1th digit - '2'.
mov ds:[bp+1],49 ; Puts in the 2th place in ID1 array the 2th digit - '1'.
mov ds:[bp+2],54 ; Puts in the 3th place in ID1 array the 3th digit - '6'.
mov ds:[bp+3],51 ; Puts in the 4th place in ID1 array the 4th digit - '3'.
mov ds:[bp+4],56 ; Puts in the 5th place in ID1 array the 5th digit - '8'.
mov ds:[bp+5],51 ; Puts in the 6th place in ID1 array the 6th digit - '3'.
mov ds:[bp+6],49 ; Puts in the 7th place in ID1 array the 7th digit - '1'.
mov ds:[bp+7],55 ; Puts in the 8th place in ID1 array the 8th digit - '7'.
mov ds:[bp+8],52 ; Puts in the 9th place in ID1 array the 9th digit - '4'.
```

ולבסוף הדפסנו למסך את הערכים:

```
; Print first ID to screen
mov bx, offset DPOINTERS ; Make sure that bx stock the offset of DPOINTERS address
mov bp, WORD [bx] ; Set bp to stock up ID1's adderess

mov al, ds:[bp] ; Set up al to stock the 1th digit of ID1
mov ah, 0Fh ; Background code: black, foreground code: white
mov es:[900h], ax ; Writing to screen memory

mov al, ds:[bp+1] ; Set up al to stock the 2th digit of ID1
mov ah, 1Eh ; Background code: blue, foreground code: yellow
mov es:[900h+2h], ax ; Writing to screen memory

mov al, ds:[bp+2] ; Set up al to stock the 3th digit of ID1
mov ah, 2Dh ; Background code: green, foreground code: light magenta
mov es:[900h+4h], ax ; Writing to screen memory

mov al, ds:[bp+3] ; Set up al to stock the 4th digit of ID1
mov ah, 3Ch ; Background code: cyan, foreground code: light red
mov es:[900h+6h], ax ; Writing to screen memory

mov al, ds:[bp+4] ; Set up al to stock the 5th digit of ID1
mov ah, 4Bh ; Background code: red, foreground code: light cyan
mov es:[900h+8h], ax ; Writing to screen memory

mov al, ds:[bp+5] ; Set up al to stock the 6th digit of ID1
mov ah, 5Ah ; Background code: magenta, foreground code: light green
mov es:[900h+8h+2h], ax ; Writing to screen memory

mov al, ds:[bp+6] ; Set up al to stock the 7th digit of ID1
mov ah, 69h ; Background code: brown, foreground code: light blue
mov es:[900h+8h+4h], ax ; Writing to screen memory

mov al, ds:[bp+7] ; Set up al to stock the 8th digit of ID1
mov ah, 78h ; Background code: light gray, foreground code: dark gray
mov es:[900h+8h+6h], ax ; Writing to screen memory

mov al, ds:[bp+8] ; Set up al to stock the 9th digit of ID1
mov ah, 12h ; Background code: blue, foreground code: green
mov es:[900h+8h+8h], ax ; Writing to screen memory

; Print secend ID to screen
mov bp, WORD [bx+2]
mov al, ds:[bp] ; Set up al to stock the 1th digit of ID1
mov ah, 19d ; Set the colors
mov es:[9A0h], ax ; Writing to screen memory

mov al, ds:[bp+1] ; Set up al to stock the 2th digit of ID1
mov ah, 28d ; Set the colors
mov es:[9A0h+2h], ax ; Writing to screen memory

mov al, ds:[bp+2] ; Set up al to stock the 3th digit of ID1
mov ah, 37d ; Set the colors
mov es:[9A0h+4h], ax ; Writing to screen memory

mov al, ds:[bp+3] ; Set up al to stock the 4th digit of ID1
mov ah, 45d ; Set the colors
mov es:[9A0h+6h], ax ; Writing to screen memory

mov al, ds:[bp+4] ; Set up al to stock the 5th digit of ID1
mov ah, 56d ; Set the colors
mov es:[9A0h+8h], ax ; Writing to screen memory

mov al, ds:[bp+5] ; Set up al to stock the 6th digit of ID1
mov ah, 64d ; Set the colors
mov es:[9A0h+8h+2h], ax ; Writing to screen memory

mov al, ds:[bp+6] ; Set up al to stock the 7th digit of ID1
mov ah, 73d ; Set the colors
mov es:[9A0h+8h+4h], ax ; Writing to screen memory

mov al, ds:[bp+7] ; Set up al to stock the 8th digit of ID1
mov ah, 82d ; Set the colors
mov es:[9A0h+8h+6h], ax ; Writing to screen memory

mov al, ds:[bp+8] ; Set up al to stock the 9th digit of ID1
mov ah, 91d ; Set the colors
mov es:[9A0h+8h+8h], ax ; Writing to screen memory
```

וניתן לראות שאכן קיבלנו על המסך את התוצאה המתבקשת:

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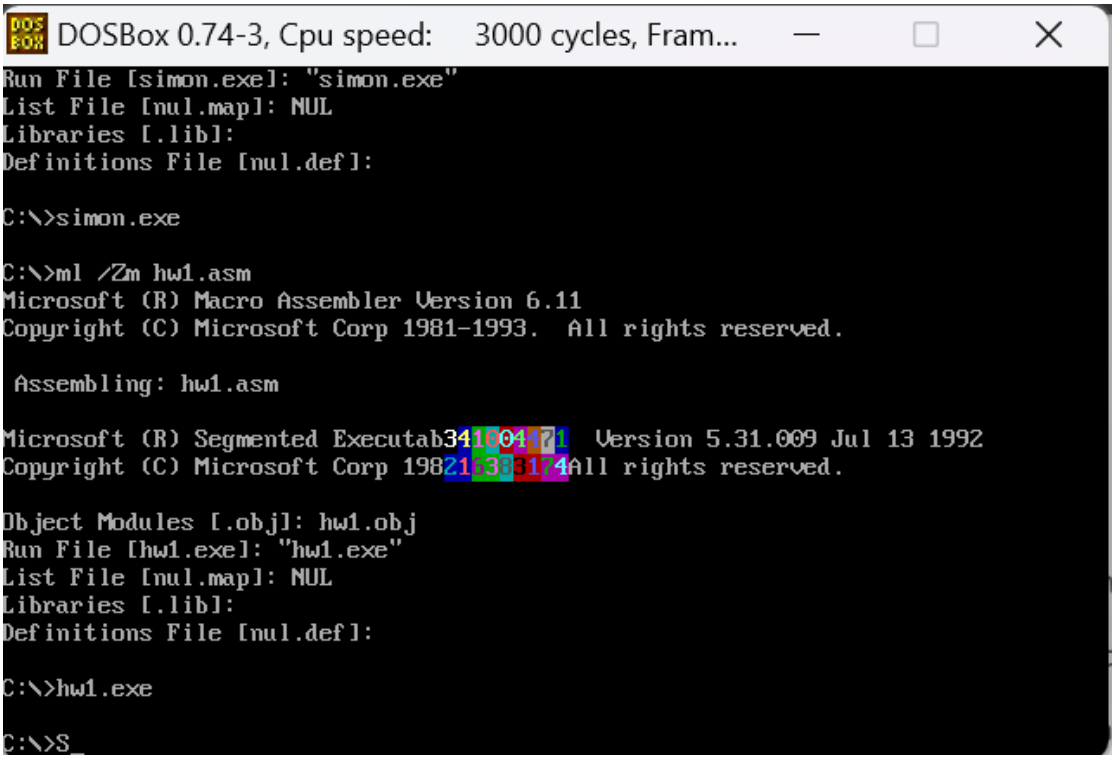
DOSBox 0.74-3, Cpu speed: 3000 cycles, Fram...
Run File [simon.exe]: "simon.exe"
List File [nul.map]: NUL
Libraries [.lib]:
Definitions File [nul.def]:
C:\>simon.exe
C:\>ml /Zm hw1.asm
Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993. All rights reserved.
Assembling: hw1.asm
Microsoft (R) Segmented Executab Version 5.31.009 Jul 13 1992
Copyright (C) Microsoft Corp All rights reserved.
Object Modules [.obj]: hw1.obj
Run File [hw1.exe]: "hw1.exe"
List File [nul.map]: NUL
Libraries [.lib]:
Definitions File [nul.def]:
C:\>hw1.exe
C:\>S